

High-Precision Tracking of Periodic Signals: A Macro–Micro Approach With Quantized Feedback

Aur lio T. Salton , Jinchuan Zheng , Senior Member, IEEE, Jeferson Vieira Flores , and Minyue Fu , Fellow, IEEE

Abstract—This article proposes a novel control design method for high-precision positioning systems. The method aims to eliminate the tracking error caused by measurement quantization present in positioning systems with incremental encoders. By employing a combined internal model based feedback and quantized feedforward design, we are able to make the output of the positioning system asymptotically track any input signal with one or more sinusoidal components of known frequencies and a possible constant component. When combined with a micro-actuator, the resulting dual-stage positioning system is able to track any continuous periodic signal with a known period. Besides theoretical guarantees, the proposed design is validated experimentally and proved able to achieve asymptotic tracking error below $\pm 1 \mu\text{m}$ when subject to a sensor quantization level of $5 \mu\text{m}$.

Index Terms—Dual-stage actuator, macro–micro, motion control, periodic signals, quantization, tracking.

I. INTRODUCTION

DESIGNERS of positioning control systems always face the tradeoff of positioning accuracy, range of actuation, and response time (or bandwidth). For example, voice coil actuators typically have a good range of actuation but their positioning accuracy is often limited by the incremental encoder and their response time is limited by the motor dynamics [1]. In contrast, piezo devices can deliver superb dynamic response due to their miniature size and nanometer accuracy due to capacitive sensing but their range of actuation is severely limited [2]–[4].

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Aur lio T. Salton and Jeferson Vieira Flores are with the School of Engineering, Universidade Federal do Rio Grande do Sul, Porto Alegre-RS 90040-060, Brazil (e-mail: aurelio.salton@ufrgs.br; jeferson.flores@ufrgs.br).

Jinchuan Zheng is with the School of Science, Computing and Engineering Technologies, Swinburne University of Technology, Hawthorn, VIC 3122, Australia (e-mail: jzheng@swin.edu.au).

Minyue Fu is with the School of Electrical Engineering and Computer Science, University of Newcastle, Callaghan, NSW 2308, Australia (e-mail: minyue.fu@newcastle.edu.au).

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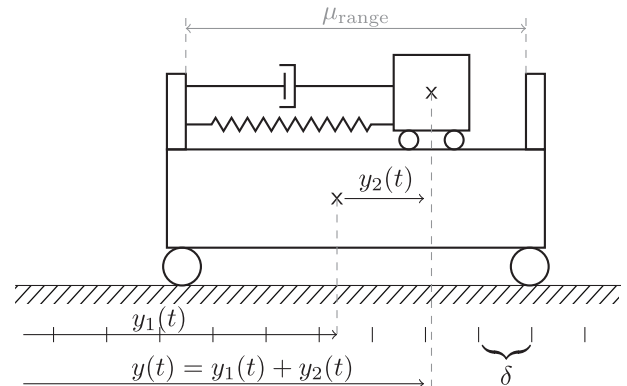


Fig. 1. Macro–micro approach: A microactuator is connected to the macroactuator (quantized) such that $y(t) = y_1(t) + y_2(t)$. Since $y_2(t)$ spans the small distance μ_{range} and is analog in nature, its own sensor quantization error is negligible [14].

To overcome this tradeoff, a dual-stage positioning system can be used. Such a system consists of a macroactuator as the primary stage and a piggy-backed microactuator as the secondary stage. Fig. 1 illustrates the schematics of such a design. Due to its relatively larger physical size, the primary stage is able to cover a long range of actuation, but with relatively slower response and coarse resolution. In contrast, the secondary stage is able to respond quickly and accurately, but only over a limited range. A typical combination for translational motion is a linear motor coupled to an incremental encoder for the primary stage, and a piezo device coupled to a capacitive sensor for the secondary stage. Through careful composite control designs, a combined system can take the advantages of both stages, providing fast and high-precision responses over a long working range. Many dual-stage designs are available in the literature; see, e.g., [5]–[8], and ample applications of dual-stage positioning systems have been demonstrated; see, e.g., [9]–[13].

Regardless whether the positioning system is single stage or multistage, the quantization error of the incremental encoder poses a fundamental challenge to the control design. This issue is particularly prominent if the system is designed for long-range high-precision tracking. In this article, we study a new control design method, which focuses on mitigating the quantization error in the feedback measurement, enabling us to significantly

improve the tracking accuracy of periodic signals way beyond the accuracy permitted by the incremental encoder. When applied in a dual-stage design for tracking periodic signals, the new control design provides the possibility to totally eliminate the tracking error caused by the incremental encoder. To do so, the macroactuator is set in a persistent motion specifically designed so that its tracking error converges to zero despite the quantization error. To track the difference between the macro-actuator motion and the actual reference, we rely on the microactuator and on its own capacitive sensor whose quantization error is intrinsically negligible due to its short range and analog nature.

Our design method was proposed and tested in our earlier paper [14] for dual-stage positioning systems to track a constant reference input, where we demonstrated how to remove the quantization effect in the primary stage. The work in this article expands this result to handle the tracking of an arbitrary periodic signal in both single-stage and dual-stage designs. The contributions and main results of this article are summarized below.

- 1) For a single-stage actuator with incremental encoder for feedback measurement, we consider the tracking control for any input signal consisting of one or more sinusoidal terms of known frequencies, including a possible constant (DC) term. We propose a novel control design consisting of internal model based feedback together with feedforward quantization and show that the controller can be carefully designed to achieve zero tracking error asymptotically. That is, the quantization effect of the incremental encoder in the feedback channel can be completely suppressed.
- 2) For tracking an arbitrary periodic signal with a known period, we deploy a dual-stage system to achieve the task. The primary stage is designed to track an approximate reference signal with a finite number of harmonic terms using the control design above. The secondary stage is designed to track the residual error. Under the assumptions that the secondary stage has negligible measurement error and sufficiently fast dynamic response (which is a reasonable assumption if the secondary stage is a piezo device with capacitive sensing), the steady-state tracking error can be made arbitrarily small for any continuous periodic input, eliminating the quantization error caused by the incremental encoder in the primary stage.
- 3) Arbitrarily small tracking errors are backed up by theoretical proofs, and the superb tracking performance is validated by experimental results. In addition, the theoretical study in the control design offers certain freedoms in the feedback controller, which can be used to tune appropriate transient response as well.

The rest of this article is organized as follows. Section II will provide preliminary material to aid the comprehension of the article. Section III will provide theoretical results that allow an output quantized actuator to asymptotically track nonstationary trajectories. These results will then be explored in Section IV, by the concept of macro–micro actuators. The proposed strategy will be illustrated by numerical and experimental results in Section V. Section VI concludes this article.

II. PROBLEM STATEMENT

Let us consider a motion system as described by a linear transfer function

$$G_1(s) := \frac{Y_1(s)}{U_1(s)} \quad (1)$$

subject to output quantization. Measurements of $y_1(t)$ are corrupted by a quantization function defined as

$$q(y_1) := j \cdot \delta \quad \forall y_1 \in \mathcal{Q}_j \quad (2)$$

where $j \in \mathbb{Z}$, $\delta \in \mathbb{R}^+$ is called the quantization interval and

$$\mathcal{Q}_j(\delta) := \{y_1 \in \mathbb{R} \mid (j - 0.5)\delta \leq y_1 < (j + 0.5)\delta\} \quad (3)$$

is the quantization region.

This article addresses the problem of tracking a periodic reference $r(t) = r(t + T)$, $T \in \mathbb{R}$, that is, given measurements $q(y_1(t))$ achieve $y_1(t) \rightarrow r(t)$ as $t \rightarrow \infty$. Two cases are considered: in the first case the reference $r(t)$ is represented by a finite Fourier series with harmonics within the bandwidth of $G_1(s)$. Then, given the feedback signal $q(y_1(t))$, the first problem to be addressed by this article is that of designing a control law such that the output $y_1(t)$ is able to asymptotically track $r(t)$.

In the second case, $r(t)$ is represented by an infinite Fourier series, and a macro–micro approach is proposed. The reference is split into $r(t) = r_1(t) + r_2(t)$, and the problem becomes that of designing a control law such that the first m harmonics of $r(t)$ are tracked by $y_1(t)$, that is, $y_1(t) \rightarrow r_1(t)$, and such that the higher order harmonics are tracked by the so-called microactuator $y_2(t) \rightarrow r_2(t)$, where

$$G_2(s) := \frac{Y_2(s)}{U_2(s)}. \quad (4)$$

A traditional macro–micro structure is depicted in Fig. 1 and arises from the addition of a microactuator to the quantized motion system—an additional sensor is also added able to measure the relative displacement $y_2(t)$. When compared to the quantized system (henceforth also called the macroactuator), the microactuator has both a smaller range and a higher bandwidth. Two advantages are obtained by this structure: 1) given its higher bandwidth, the microactuator is able to track the higher order harmonics $r_2(t)$ in $r(t)$, and 2) given its reduced range (μ_{range} in Fig. 1), measurement errors of $y_2(t)$ are negligible when compared to quantization ones $e_q(t) = q(y_1(t)) - y_1(t)$. As such, even if asymptotic tracking of $r_2(t)$ may be impossible for some references, a significantly improved accuracy is achieved by the macro–micro structure provided the higher harmonics in $r_2(t)$ are within the micro-actuator range.

The overall strategy is depicted in Fig. 2, where as in Fig. 1, it is clear that the total output is now given by $y(t) = y_1(t) + y_2(t)$. The periodic reference $r(t)$ is accordingly split as $r(t) = r_1(t) + r_2(t)$, and two quantization blocks are present: an intrinsic one in the feedback path of $y_1(t)$, and the artificial one in the feedforward path of $r_1(t)$. We will now proceed by focusing on the tracking control problem for the macroactuator and later show how to include the micro one in order to achieve an improved performance.

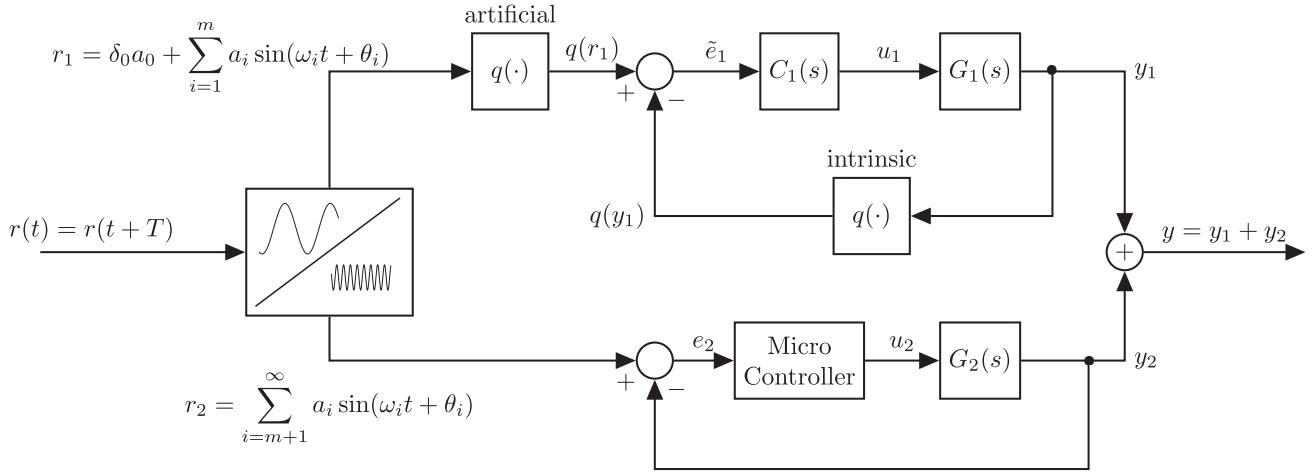


Fig. 2. Schematic representation of the proposed strategy: The reference $r_1(t)$ is artificially quantized and asymptotically tracked; the microactuator is then used to compensate for the residual signal $r_2(t) = r(t) - r_1(t)$.

III. MACRO-STAGE DESIGN

Let the input signal $r_1(t)$ be of the following form:

$$r_1(t) = \delta_0 a_0 + \sum_{i=1}^m a_i \sin(\omega_i t + \theta_i) \quad (5)$$

where $\delta_0 = 0$ or 1 , $m \geq 1$, $\omega_i > 0$, $\theta_i \in \mathbb{R}$ and $a_i \neq 0$ for all i . Note, in particular, that $r_1(t)$ contains at least one sinusoid.

Remark 1: The above structure for $r_1(t)$ allows the given reference signal to be one or more sinusoids dislocated by a_0 . This permits good approximation of any periodic reference signal by having a sufficient number of sinusoids. As we will see later in the dual-stage design, for any given periodic reference input signal, (5) can be used as the approximated reference input for the primary stage, leaving the small approximation error to be handled by the secondary stage. We also note that (5) also allows quasi-periodic reference signals, where $\omega_i, i > 1$, are not necessarily the harmonics of ω_1 .

Assumption 1: The magnitude range of $r_1(t)$ is sufficiently large with respect to the quantization step size Δ such that $q(r_1(t))$ has at least $2m + \delta_0$ different quantization values.

This assumption requires that two quantization levels be crossed for each sinusoidal component in the reference $r_1(t)$, and an extra level if $r_1(t)$ has a DC component.

Lemma 1: Under Assumption 1, $r_1(t)$ is unique in the sense that for any $\tilde{r}_1(t)$ of the form

$$\tilde{r}_1(t) = \delta_0 \tilde{a}_0 + \sum_{i=1}^m \tilde{a}_i \sin(\omega_i t + \tilde{\theta}_i)$$

with $q(\tilde{r}_1(t)) = q(r_1(t))$ for all t , then, $\tilde{r}_1(t) = r_1(t)$ for all t .

Proof: It is easy to see that $r_1(t)$ in (5) has only $2m + \delta_0$ free parameters (i.e., all a_i and θ_i). Therefore, it is not possible to have two different functions of the same form with the identical at least $2m + \delta_0$ different values. ■

Consider the open-loop transfer function $H(s) = C(s)G(s)$ from $\tilde{e}_1(t) := q(r_1(t)) - q(y_1(t))$ to $y_1(t)$ as given by

$$H(s) = \delta_0 \frac{k_0}{s} + \sum_{i=1}^m \frac{k_i s}{s^2 + \omega_i^2} + \sum_{i=m+1}^{m+p} \frac{k_i}{s + p_i} + \sum_{i=m+p+1}^{m+p+q} \frac{k_i (s + z_i)}{s^2 + c_i s + d_i}. \quad (6)$$

Assumption 2: Suppose the following conditions hold:

- 1) $k_i > 0$ for all $i = 0, 1, \dots, m + p + q$.
- 2) $p \geq 0$; $p_i > 0$ for all $i = m + 1, \dots, m + p$.
- 3) $q \geq 0$; $c_i > 0, d_i > 0$ and $0 \leq z_i \leq c_i$ for all $i = m + p + 1, \dots, m + p + q$.

The terms $i = 0, \dots, m$ are necessary in order to match the reference and satisfy the internal model principle: $H_1(s)$ includes an integrator for DC tracking, and a pair of complex poles for each sinusoid in the signal to be tracked, thus matching the reference model. The first-order terms ($i = m + 1, \dots, m + p$) and second-order terms ($i = m + p + 1, \dots, m + p + q$) are strictly Hurwitz, but they may be void (i.e., $p = q = 0$) because, as we see below, their presence does not influence the asymptotic behavior of the closed-loop system. However, the inclusion of these terms can be beneficial for shaping the transient response, offering some design flexibility, and accommodating the plant dynamics.

We have the following main result.

Theorem 1: Suppose $r_1(t)$ and $H(s)$ are as in (5) and (6), respectively, and that Assumptions 1 and 2 hold. Then, the closed-loop system is stable and the tracking error $e_1(t) = r_1(t) - y_1(t) \rightarrow 0$ as $t \rightarrow \infty$.

Proof: Define the input components as

$$r_{1,0}(t) = a_0$$

$$r_{1,i}(t) = a_i \sin(\omega_i t + \theta_i), i = 1, \dots, m$$

and the open-loop subsystems as

$$H_{1,0}(s) = \frac{k_0}{s}$$

$$H_{1,i}(s) = \frac{k_i s}{s^2 + \omega_i^2}, i = 1, \dots, m$$

$$H_{1,i}(s) = \frac{k_i}{s + p_i}, i = m + 1, \dots, m + p$$

$$H_{1,i}(s) = \frac{k_i(s + z_i)}{s^2 + c_i s + d_i}, i = m + p + 1, \dots, m + p + q.$$

Let the output of each subsystem be $y_{1,i}(t)$, $i = 1, \dots, m + p + q$. We have

$$\dot{y}_{1,0}(t) = k_0 \tilde{e}_1(t)$$

$$\ddot{y}_{1,i}(t) + \omega_i^2 y_{1,i}(t) = k_i \dot{\tilde{e}}_1(t), i = 1, \dots, m$$

$$\dot{y}_{1,i}(t) + p_i y_{1,i}(t) = k_i \tilde{e}_1(t)$$

$$i = m + 1, \dots, m + p$$

$$\ddot{y}_{1,i}(t) + c_i \dot{y}_{1,i}(t) + d_i y_{1,i}(t) = k_i (\dot{\tilde{e}}_1(t) + z_i \tilde{e}_1(t))$$

for $i = m + p + 1, \dots, m + p + q$.

It is easy to verify that

$$\dot{r}_{1,0}(t) = 0$$

$$\ddot{r}_{1,i}(t) + \omega_i^2 r_{1,i}(t) = 0, i = 1, \dots, m.$$

We also define

$$r_{1,i}(t) = 0, i = m + 1, \dots, m + p + q$$

$$e_{1,i}(t) = r_{1,i}(t) - y_{1,i}(t), i = 0, 1, \dots, m + p + q.$$

It follows that

$$\dot{e}_{1,0}(t) = -k_0 \tilde{e}_1(t) \quad (7)$$

$$\ddot{e}_{1,i}(t) + \omega_i^2 e_{1,i}(t) = -k_i \dot{\tilde{e}}_1(t), i = 1, \dots, m \quad (8)$$

$$\dot{e}_{1,i}(t) + p_i e_{1,i}(t) = -k_i \tilde{e}_1(t) \quad (9)$$

$$i = m + 1, \dots, m + p$$

$$\ddot{e}_{1,i}(t) + c_i \dot{e}_{1,i}(t) + d_i e_{1,i}(t) = -k_i (\dot{\tilde{e}}_1(t) + z_i \tilde{e}_1(t))$$

$$i = m + p + 1, \dots, m + p + q. \quad (10)$$

Moreover, it is straightforward to verify that

$$e_1(t) = \delta_0 e_{1,0}(t) + \sum_{i=1}^{m+p+q} e_{1,i}(t). \quad (11)$$

Now construct the Lyapunov-like function

$$V(t) = \sum_{i=0}^{m+p+q} \frac{1}{k_i} V_i(t) > 0 \quad (12)$$

where each $V_i(t)$ is for each subsystem in (7)–(10), as defined and analyzed below.

Case 0: $i = 0$. For this case, we take

$$V_0(t) = \frac{\delta_0}{2} (e_{1,0}(t))^2. \quad (13)$$

Taking derivative, we get

$$\dot{V}_0(t) = \delta_0 \dot{e}_{1,0}(t) e_{1,0}(t) = -k_0 \delta_0 e_{1,0}(t) \tilde{e}_1(t).$$

Case 1: $i = 1, \dots, m$. For this case, we take

$$V_i(t) = \frac{1}{2} (e_{1,i}(t))^2 + \frac{1}{2\omega_i^2} (\dot{e}_{1,i}(t) + k_i \tilde{e}_1(t))^2. \quad (14)$$

Similarly, taking derivative, we get

$$\begin{aligned} \dot{V}_i(t) &= \dot{e}_{1,i}(t) e_{1,i}(t) + \frac{1}{\omega_i^2} (\ddot{e}_{1,i}(t) + k_i \dot{\tilde{e}}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + k_i \tilde{e}_1(t)) \\ &= \dot{e}_{1,i}(t) e_{1,i}(t) - e_{1,i}(t) (\dot{e}_{1,i}(t) + k_i \tilde{e}_1(t)) \\ &= -k_i e_{1,i}(t) \tilde{e}_1(t). \end{aligned}$$

Case 2: $i = m + 1, \dots, m + p$. For this case, we take

$$V_i(t) = \frac{1}{2} (e_{1,i}(t))^2. \quad (15)$$

Again, taking derivative, we get

$$\begin{aligned} \dot{V}_i(t) &= \dot{e}_{1,i}(t) e_{1,i}(t) \\ &= (-p_i e_{1,i}(t) - k_i \tilde{e}_1(t)) e_{1,i}(t) \\ &= -p_i (e_{1,i}(t))^2 - \tilde{e}_1(t) e_{1,i}(t) \\ &\leq -k_i e_{1,i}(t) \tilde{e}_1(t). \end{aligned}$$

Case 3: $i = m + p + 1, \dots, m + p + q$. This is the most complicated case. We consider several subcases. Denote by $-p_{i1}$ and $-p_{i2}$ the two poles of $H_{1,i}(s)$, i.e., the denominator of $H_{1,i}(s)$ is expressed as $(s + p_{i1})(s + p_{i2})$. We have $c_i = p_{i1} + p_{i2}$ and $d_i = p_{i1} p_{i2}$. By Assumption 2, $0 \leq z_i \leq p_{i1} + p_{i2}$.

Case 3.1: The poles are real and $H_{1,i}(s)$ has zero-pole cancellation, i.e., $p_{i1} > 0, p_{i2} > 0$ and $z = p_{i1}$ or p_{i2} . In this subcase, the simplified $H_{1,i}(s) = k_i/(s + p_{i1})$ or $k_i/(s + p_{i2})$. This is the same as Case 2. So we take $V_i(t)$ as in (15) which yields

$$\dot{V}_i(t) \leq -k_i e_{1,i}(t) \tilde{e}_1(t).$$

Case 3.2: The poles are real and $0 < p_{i1} < z < p_{i2}$. Taking partial fraction expansion, $H_{1,i}(s)$ can be rewritten as

$$H_{1,i}(s) = \frac{k_{i1}}{s + p_{i1}} + \frac{k_{i2}}{s + p_{i2}}.$$

It is straightforward to check that

$$k_{i1} + k_{i2} = k_i, \quad \frac{k_{i1}}{k_i} p_{i2} + \frac{k_{i2}}{k_i} p_{i1} = z.$$

Defining $\varepsilon_i = k_{i1}/k_i$, we get

$$\varepsilon_i (p_{i2} - p_{i1}) + p_{i1} = z.$$

Note that the left-hand side above is linear in ε . Hence, $p_{i1} < z_i < p_{i2}$ implies that $0 < \varepsilon_i < 1$, which in turn implies that both $k_{i1} > 0$ and $k_{i2} > 0$. Therefore, this subcase goes back to Case 2. We have $e_{1,i}(t) = e_{i1}(t) + e_{i2}(t)$, where

$$\dot{e}_{ij}(t) + p_{ij} e_{ij}(t) = -k_{ij} \tilde{e}(t), \quad j = 1, 2.$$

Take

$$V_i(t) = \frac{k_i}{k_{i1}}(e_{i1}(t))^2 + \frac{k_i}{k_{i2}}(e_{i2}(t))^2. \quad (16)$$

From Case 2, we get

$$\begin{aligned} \dot{V}_i(t) &\leq -\frac{k_i}{k_{i1}}k_{i1}e_{i1}(t)\tilde{e}_1 - \frac{k_i}{k_{i2}}k_{i2}e_{i2}(t)\tilde{e}(t) \\ &= -k_ie_{1,i}(t)\tilde{e}_1(t). \end{aligned}$$

Case 3.3: The poles are real and either $0 \leq z_i < p_{i1} \leq p_{i2}$ or $p_{i2} < z_i \leq c_i$. Note that this subcase includes the special case of repeated poles, where $p_{i1} = p_{i2}$ and either $0 \leq z_i < p_{i1}$ or $p_{i1} < z_i \leq 2p_{i1}$.

Define

$$\begin{aligned} \rho_i &= d_i - (c_i - z_i)z_i \\ &= p_{i1}p_{i2} - (p_{i1} + p_{i2} - z_i)z_i \\ &= (z - p_{i1})(z - p_{i2}). \end{aligned}$$

Hence, $\rho_i > 0$ by the conditions of this subcase. Take

$$\begin{aligned} V_i(t) &= \frac{1}{2}(e_{1,i}(t))^2 \\ &\quad + \frac{1}{2\rho_i}(\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2. \end{aligned} \quad (17)$$

We have

$$\begin{aligned} \dot{V}_i(t) &= e_{1,i}(t)\dot{e}_{1,i}(t) + \frac{1}{\rho_i}(\ddot{e}_{1,i}(t) + (c_i - z_i)\dot{e}_{1,i}(t) \\ &\quad + k_i\dot{\tilde{e}}_1(t))(\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= e_{1,i}(t)\dot{e}_{1,i}(t) + \frac{1}{\rho_i}(-z_i\dot{e}_{1,i}(t) - d_ie_{1,i}(t) - k_iz_i\tilde{e}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= e_{1,i}(t)\dot{e}_{1,i}(t) \\ &\quad - \frac{1}{\rho_i}(z_i\dot{e}_{1,i}(t) + (c_i - z_i)z_ie_{1,i}(t) + z_ik_i\tilde{e}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &\quad - \frac{d_i - (c_i - z_i)z_i}{\rho_i}e_{1,i}(t) \\ &\quad \times (\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= e_{1,i}(t)\dot{e}_{1,i}(t) \\ &\quad - \frac{z_i}{\rho_i}(\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2 \\ &\quad - e_{1,i}(t)(\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= -\frac{z_i}{\rho_i}(\dot{e}_{1,i}(t) + (c_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2 \\ &\quad - (c_i - z_i)(e_{1,i}(t))^2 - k_ie_{1,i}(t)\tilde{e}_1(t) \\ &\leq -k_ie_{1,i}(t)\tilde{e}_1(t). \end{aligned}$$

Case 3.4: Complex poles; $p_{i1} = \alpha_i + j\beta_i$, $p_{i2} = \alpha_i - j\beta_i$ with $\alpha_i > 0$ and $\beta_i > 0$. We assume $\beta_i > 0$ because the special case of $\beta_i = 0$ corresponds to the repeated pole case, which has

been considered in Case 3.3. Rewrite (10) as

$$\ddot{e}_{1,i}(t) + 2\alpha_i\dot{e}_{1,i}(t) + \gamma_i^2 e_{1,i}(t) = -k_i(\dot{\tilde{e}}_1(t) + z_i\tilde{e}_1(t))$$

where $\gamma_i^2 = \alpha_i^2 + \beta_i^2$. Note that $0 \leq z_i \leq c_i = 2\alpha_i$ by Assumption 2. Take

$$\begin{aligned} V_i(t) &= \frac{1}{2}(e_{1,i}(t))^2 \\ &\quad + \frac{1}{2\eta_i^2}(\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2 \end{aligned} \quad (18)$$

where $\eta_i^2 = (z_i - \alpha_i)^2 + \beta_i^2$. Note that $\eta_i^2 > 0$ because $\beta_i > 0$. Taking the derivative, we get

$$\begin{aligned} \dot{V}_i(t) &= \dot{e}_{1,i}(t)e_{1,i}(t) + \frac{1}{\eta_i^2}(\ddot{e}_{1,i}(t) + (2\alpha_i - z_i)\dot{e}_{1,i}(t) + k_i\dot{\tilde{e}}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= \dot{e}_{1,i}(t)e_{1,i}(t) + \frac{1}{\eta_i^2}(-z_i\dot{e}_{1,i}(t) - \gamma_i^2 e_{1,i}(t) - k_iz_i\tilde{e}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= \dot{e}_{1,i}(t)e_{1,i}(t) \\ &\quad + \frac{1}{\eta_i^2}(-z_i\dot{e}_{1,i}(t) - (2\alpha_i - z_i)z_ie_{1,i}(t) - k_iz_i\tilde{e}_1(t)) \\ &\quad \times (\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &\quad - \frac{\gamma_i^2 - (2\alpha_i - z_i)z_i}{\eta_i^2}e_{1,i}(t)\xi(t) \end{aligned}$$

with $\xi(t) = (\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))$. It is easy to verify that

$$\begin{aligned} \gamma_i^2 - (2\alpha_i - z_i)z_i &= \alpha_i^2 - (2\alpha_i - z_i)z_i + \beta_i^2 \\ &= (z_i - \alpha_i)^2 + \beta_i^2 \\ &= \eta_i^2. \end{aligned}$$

We have

$$\begin{aligned} \dot{V}_i(t) &= \dot{e}_{1,i}(t)e_{1,i}(t) \\ &\quad - \frac{z_i}{\eta_i^2}(\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2 \\ &\quad - e_{1,i}(t)(\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t)) \\ &= -\frac{z_i}{\eta_i^2}(\dot{e}_{1,i}(t) + (2\alpha_i - z_i)e_{1,i}(t) + k_i\tilde{e}_1(t))^2 \\ &\quad - (2\alpha_i - z_i)(e_{1,i}(t))^2 - k_ie_{1,i}(t)\tilde{e}_1(t) \\ &\leq -k_ie_{1,i}(t)\tilde{e}_1(t). \end{aligned}$$

Combining all the cases above, we get

$$\begin{aligned} \dot{V}(t) &= \sum_{i=0}^{m+p+q} \frac{1}{k_i} \dot{V}_i(t) \\ &\leq -\left(\delta_0 e_{i,0}(t) + \sum_{i=1}^{m+p+q} e_{1,i}(t)\right) \tilde{e}_1(t) \\ &= -e_1(t)\tilde{e}_1(t). \end{aligned}$$

Note that $e_1(t)\tilde{e}_1(t) \geq 0$ because the quantization function is monotonic (i.e., $r_1 - y_1 \geq 0 \Rightarrow q(r_1) - q(y_1) \geq 0$). This implies that, in a steady state, $V(t)$ must be confined in the region, where $e_1(t)\tilde{e}_1(t) = 0$, i.e., $e_1(t) = 0$ or $\tilde{e}_1(t) = 0$. Also note that $e_1(t) = 0$ implies $\tilde{e}_1(t) = 0$. Therefore, this region is equivalent to $\tilde{e}_1(t) = q(r_1(t)) - q(y_1(t)) = 0$.

Now we focus on the steady state of the output $y_1(t)$, which is given by the persistent terms in $H_1(s)$. From Assumption 2, it follows that

$$y_1(t) = \delta_0 y_{1,0} + \sum_{i=1}^m y_{1,i} \sin(\omega_i t + \psi_i).$$

Note that this is the same form as $r_1(t)$. By Lemma 1, $\tilde{e}_1(t) = q(r_1(t)) - q(y_1(t)) = 0$ for all t implies that $y_1(t) = r_1(t)$ in a steady state. That is, the tracking error $e(t) \rightarrow 0$ as $t \rightarrow \infty$. ■

IV. DUAL-STAGE DESIGN

The dual-stage design is illustrated in Fig. 2, where the macro- and microactuators are respectively depicted as $G_1(s)$ and $G_2(s)$.

Instead of tracking the desired periodic reference $r(t)$, the macroactuator will be set in a persistent motion around the reference in a trajectory $r_1(t)$ in the form of (5). That is, $r_1(t)$ is a periodic signal that consists of at least one sinusoidal term—and a possible dc term—that approximates $r(t)$. The approximation is done such that the residual signal $r(t) - r_1(t)$ can be tracked by the secondary stage. In the case that the reference signal $r(t)$ is a constant (i.e., dc only), we refer back to the idea in our earlier paper [14], which makes $r_1(t) = r(t) + a_1 \sin(\omega_1 t)$ with a relatively small a_1 so that $r_1(t)$ has a sinusoidal term and the residual signal $r(t) - r_1(t)$ can be tracked by the secondary stage as well.

With the above configuration, our proposed control design for the primary stage in the previous section can be applied to ensure asymptotic tracking of $r_1(t)$ despite sensor quantization. For a given plant $G_1(s)$, it suffices to design the controller $C_1(s)$ such that $H_1(s) = C_1(s)G_1(s)$ is in the form of (6). If designing such $C_1(s)$ is not possible (for example, if $G_1(s)$ has marginal or unstable poles, or if a noncausal controller would be required to match the relative degree of $H_1(s)$), then design $C_1(s)$ such that it is a stabilizing controller and $H_1(j\omega)$ approximates that in (6) as much as possible for the frequency range of interest. In this case, there may be some small steady-state error.

Since the overall output of the system is now given by $y(t) = y_1(t) + y_2(t)$, the task of the microactuator is to compensate for the residual signal by means of

$$r_2(t) = r(t) - r_1(t) \quad (19)$$

from which it follows that $y(t)$ will asymptotically track $r(t)$ if each actuator asymptotically tracks its own reference. The secondary actuator tracking of $r_2(t)$ is a standard control problem that may be addressed by traditional methods based on the internal model principal such as, e.g., resonant or repetitive controllers [15], iterative learning [16].

Remark 2: The splitting of $r(t)$ into $r_1(t)$ and $r_2(t)$ must take into account the limitations of both actuators: while $r_1(t)$ must

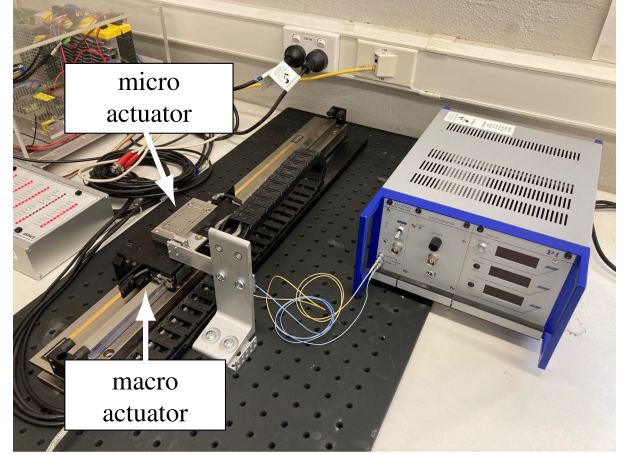


Fig. 3. Experimental macro-micro actuator comprised of a linear motor and a piezoelectric device [8].

respect Assumption 1, $r_2(t)$ must respect the range limitation of the secondary actuator.

V. EXPERIMENTAL RESULTS

The experimental setup used for the validation of the proposed results is comprised of a linear motor and a piezoelectric device, as depicted in Fig. 3, and controlled by a DSP system (dSPACE-DS1103) working at 5 kHz. Both outputs $y_1(t)$ and $y_2(t)$ are measured by capacitive sensors with nanometer resolution but, in order to illustrate the contribution, an artificial quantization level of $\delta = 5 \mu\text{m}$ was applied to $y_1(t)$ when used for feedback. This setup was identified in [8] with the macroactuator model described by a mass subject to damping

$$G_1(s) = \frac{k}{s(s+a)} \quad (20)$$

with $k = 1.7 \times 10^7$ and $a = 9.42$, and the micro-actuator model given by

$$G_2(s) = \frac{3.032 \times 10^6}{s^2 + 1810s + 1.011 \times 10^6}. \quad (21)$$

Triangular and trapezoidal waves with fundamental frequency $\omega_r = 2\pi \text{ rad/s}$ described in Fig. 4 will be assigned to $r(t)$, both of which have half wave symmetry and are solely composed of odd harmonics. We will design $C_1(s)$ according to (6), so that it tracks the fundamental ω_r and the first odd harmonic $3\omega_r$ of the reference $r(t)$. Therefore, the desired $H_1(s) = C_1(s)G_1(s)$ must encompass the poles of $G_1(s)$ and two extra pairs of complex poles

$$\begin{aligned} H_1(s) &= \frac{k_0}{s} + \frac{k_1 s}{s^2 + \omega_r^2} + \frac{k_2 s}{s^2 + (3\omega_r)^2} + \frac{k_3}{s+a} \\ &= \frac{N_H(s)}{s(s^2 + \omega_r^2)(s^2 + (3\omega_r)^2)(s+a)} \end{aligned} \quad (22)$$

where $N_H(s)$ is a fifth degree polynomial

$$N_H(s) = A_5 s^5 + A_4 s^4 + A_3 s^3 + A_2 s^2 + A_1 s + A_0. \quad (23)$$

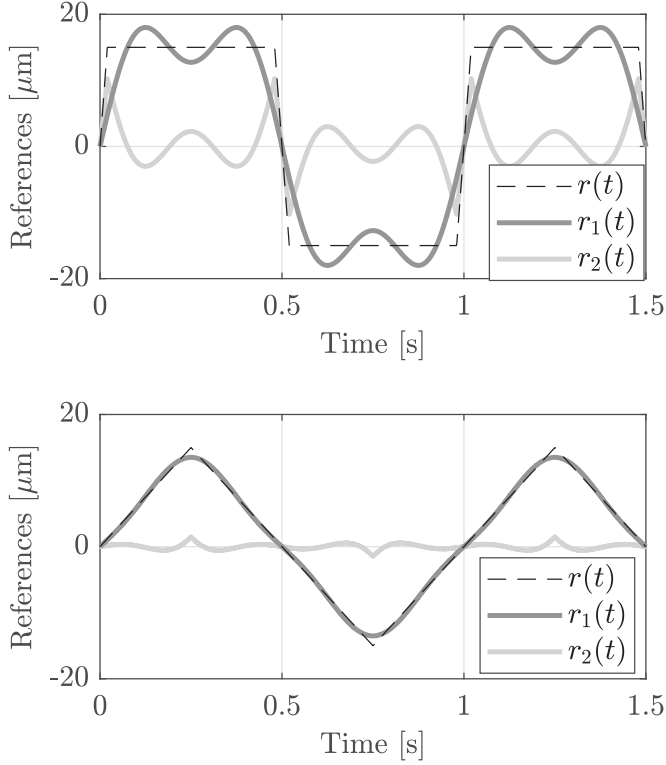


Fig. 4. References considered for tracking: $r_1(t)$ is defined as the first two harmonics of $r(t)$, the remaining harmonics are left for $r_2(t) = r(t) - r_1(t)$.

TABLE I
CONTROL PARAMETERS FOR $C_1(s)$

k_0	k_1	k_2	k_3	k_{bw}	$p_{\bar{\omega}}$
1	10	10	0.1	10	10^3

In order to satisfy Assumption 2, it follows that $k_i > 0$, $i = 0, \dots, 3$ and consequently $A_j > 0$, $j = 0, \dots, 5$. Therefore, (22) has relative degree one and, since $G_1(s)$ has relative degree two, a noncausal controller would be required to satisfy $C_1(s)G_1(s) = H_1(s)$. To avoid this problem, an extra pole $p_{\bar{\omega}}$ is added to $C_1(s)$ which, together with $k_0, \dots, k_3$, is chosen so that the closed loop is stable and (22) is satisfied for a large frequency range

$$C_1(s) = \frac{k_{bw}}{k} \frac{p_{\bar{\omega}}}{s + p_{\bar{\omega}}} \frac{N_H(s)}{(s^2 + \omega_r^2)(s^2 + (3\omega_r)^2)}.$$

In the above, k_{bw} and $p_{\bar{\omega}} > (3\omega_r)^2$ are free parameters that respectively set the bandwidth of $C_1(s)G_1(s)$ and determine the frequency up to which (22) will be valid. These parameters are listed in Table I. Since the control strategy is implemented in discrete time with $T = 0.2$ ms, the following discrete control (obtained by the matched discretization method) was implemented:

$$C_1(z) = 10^{-3} \times \frac{z^5 - 4.99z^4 + 9.99z^3 - 9.99z^2 + 4.99z - 0.998}{z^5 - 4.81z^4 + 9.27z^3 - 8.91z^2 + 4.27z - 0.818}. \quad (24)$$

The microactuator is left with the task of tracking the remaining harmonics present in $r(t)$. Since triangular and trapezoidal signals have infinite harmonics, so does $r_2(t) = r(t) - r_1(t)$, making the repetitive controller strategy a suitable candidate for

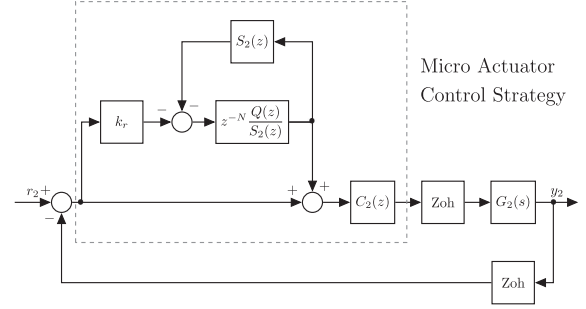


Fig. 5. Plug-in repetitive control strategy applied to the microactuator micro.

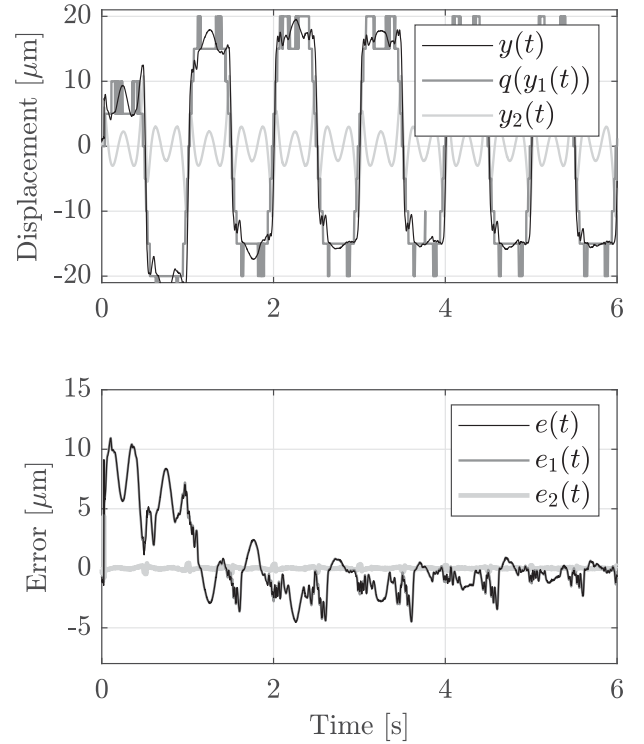


Fig. 6. Experimental results: Tracking of a trapezoidal wave. Top plot (outputs): total $y(t) = y_1(t) + y_2(t)$, quantized $q(y_1(t))$ and microactuator $y_2(t)$. Bottom plot (errors): total $e(t)$, macro $e_1(t) = r_1(t) - y_1(t)$, and micro $e_2(t) = r_2(t) - y_2(t)$.

the microloop. Furthermore, because this actuator is lighter and much faster (higher bandwidth) than the macro one, its effect on the overall settling time is minimal and his controller may indeed focus on the steady-state performance. Thus, we have opted for a zero-phase discrete-time odd-harmonic repetitive controller, as depicted in Fig. 5, and followed the design steps in [17]. Since hysteresis is typically not a significant issue in this plant [8], it is assumed to be negligible and its discrete-time control law was designed based on traditional loop shaping techniques, leading to

$$C_2(z) = 0.25 \frac{z^2 - 1.653z + 0.6896}{(z-1)(z-0.8187)} \quad (25)$$

and applied together with a zero-phase filter $Q(z) = 0.44z + 0.44 + 0.11z^{-1}$. The remaining parameters in Fig. 5 are $S_2(z) = C_2(z)G_2(z)/(1 + C_2(z)G_2(z))$, $k_r = 1.2$ and $N = 2500$.

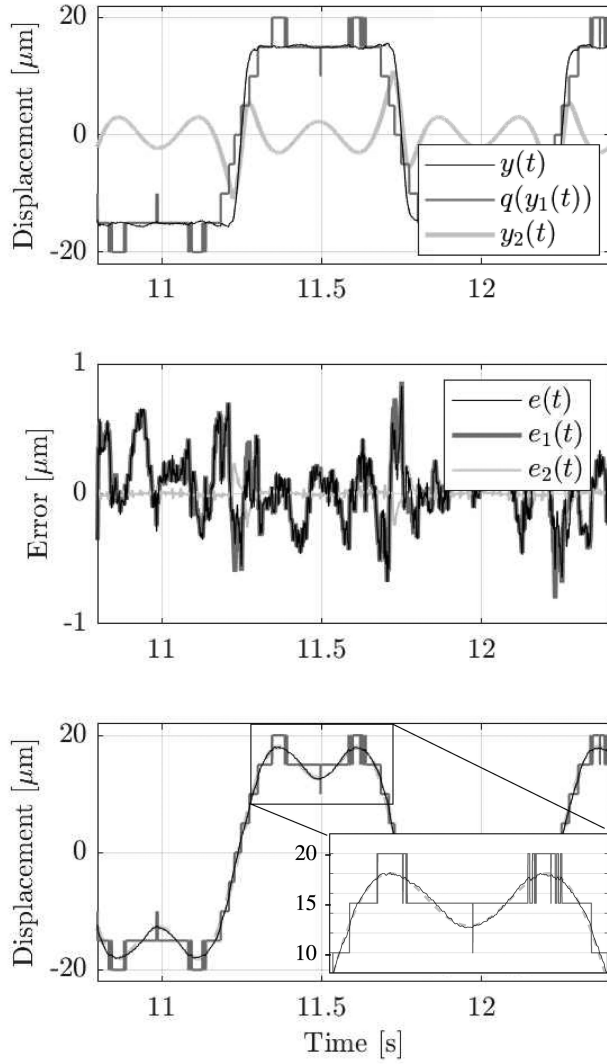


Fig. 7. Experimental results: Tracking of a trapezoidal wave. Top and middle plot are as before. Bottom plot: reference $r_1(t)$, quantized output $q(y_1(t))$ used for feedback, and the actual position of $y_1(t)$. The overall tracking error is well within the $\pm 1 \mu\text{m}$ range, despite the sensor quantization of $\delta = 5 \mu\text{m}$.

A. Tracking a Trapezoidal Wave

Here we show the tracking of a symmetric Trapezoidal wave with amplitude $A_{tr} = 15 \mu\text{m}$ and slope $a_s = 0.875 \mu\text{m/ms}$. The macroactuator is set to track the first two harmonics

$$r_1(t) = \frac{4A_{tr}}{\pi} \left(\sin(\omega_r t) + \frac{1}{3} \sin(3\omega_r t) \right) \quad (26)$$

and the microactuator tracks the difference $r_2(t) = r(t) - r_1(t)$. The respective reference signals are shown in the top plot of Fig. 4, transient results are depicted in Fig. 6 and steady-state results are depicted in Fig. 7. The bottom plot of Fig. 6 shows that the tracking error converges in about five reference cycles, and the middle plot of Fig. 7 shows the steady-state tracking error. The following approximations in $C_1(s)$ and $G_1(s)$ resulted in $C_1(s)G_1(s) \approx H_1(s)$ and prevented the error from converging to zero: 1) the presence of noise and nonlinear phenomena—such as backlash and static friction—are not captured by the ideal model $G_1(s)$; and 2) controller $C_1(s)$ required the additional

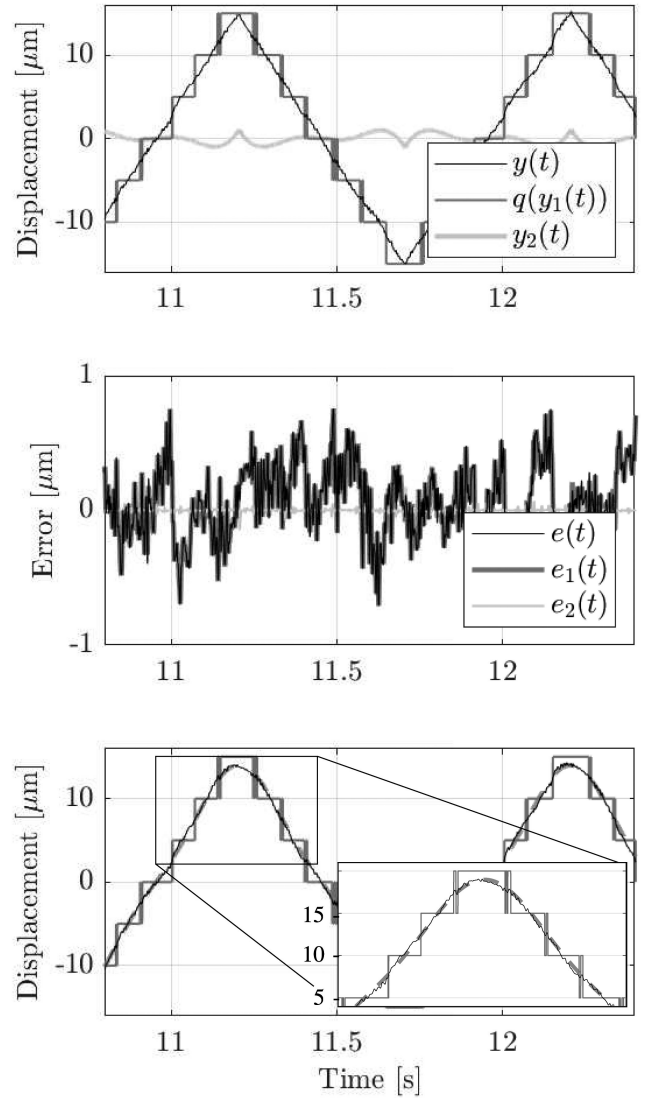


Fig. 8. Experimental results: Tracking of a triangular wave. Top and middle plot are as before. Bottom plot: reference $r_1(t)$, the quantized output $q(y_1(t))$ used for feedback, and the actual position of $y_1(t)$. The overall tracking error is well within the $\pm 1 \mu\text{m}$ range, despite the sensor quantization of $\delta = 5 \mu\text{m}$.

pole at $p_{\bar{\omega}}$ in order to achieve causality and was implemented in discrete time with a large sampling time of $T = 0.2 \text{ ms}$. Nevertheless, the resulting error is below the $\pm 1 \mu\text{m}$ mark, despite sensor quantization of $\delta = 5 \mu\text{m}$.

B. Tracking a Triangular Wave

The next signal to be tracked is a triangular wave with amplitude $A_{tr} = 15 \mu\text{m}$ and, as before, $\omega_r = 2\pi \text{ rad/s}$. The first two harmonics, set as the macro-actuator reference, are now given by

$$r_1(t) = \frac{8A_{tr}}{\pi^2} \left(\sin(\omega_r t) - \frac{1}{9} \sin(3\omega_r t) \right). \quad (27)$$

Reference signals are shown in the bottom plot of Fig. 4 and tracking results are depicted in Fig. 8. Very similar results to those obtained by the trapezoidal signal are shown. There

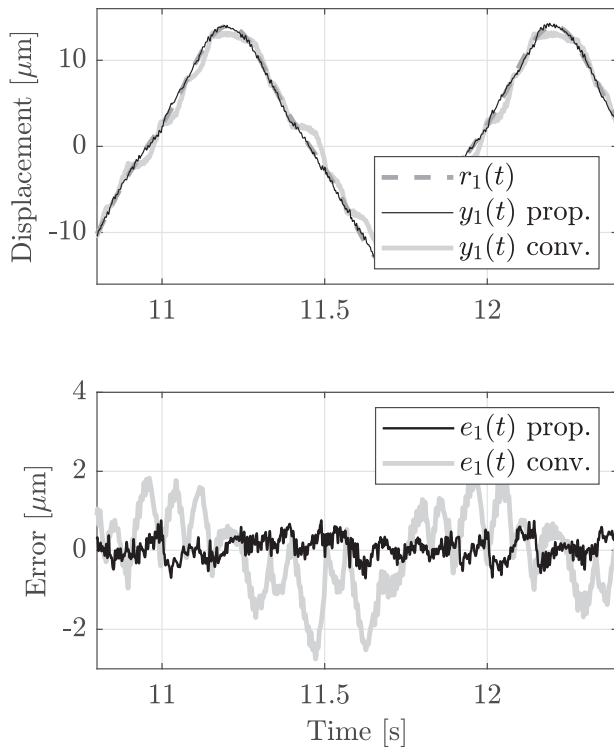


Fig. 9. Experimental results: Proposed and conventional methods compared when tracking a triangular wave. Top plot: reference $r_1(t)$, proposed tracking method “ $y_1(t)$ prop.” and conventional method “ $y_1(t)$ conv.” Bottom plot (macroerrors): proposed “ $e_1(t)$ prop.” and conventional “ $e_1(t)$ conv.”

is a residual error due to approximations in $C_1(s)$ and due to nonlinear phenomena and noise. Nevertheless, the overall experimental error is well below the $\pm 1 \mu\text{m}$ mark, despite sensor quantization of $\delta = 5 \mu\text{m}$.

The next plots, depicted in Fig. 9, compare the proposed approach to a conventional one. The same trajectories and controllers are implemented, but the artificial quantization block in reference $r_1(t)$ is removed. As a consequence, the effects of output quantization are clearly seen on the tracking performance of $y_1(t)$, and a larger steady-state error is present.

VI. CONCLUSION

A novel control design method for high-precision positioning systems to counter the quantization errors in the feedback channel was presented. In theory, the method was able to eliminate the tracking error for any periodic reference input with a known period. It was also validated experimentally that the tracking error can be suppressed down to the level substantially lower than the quantization error. The proposed method was effective for both single-stage actuators and dual-stage actuators. Future work can aim at generalizing the method to handle a broader class of systems, a wider range of tracking problems, and some types of measurement uncertainties, such as AD conversion.

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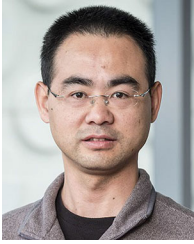
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Aurélio T. Salton received the B.Eng. degree in control and automation engineering from the Pontifical Catholic University, Rio Grande do Sul, Porto Alegre, Brazil, in 2007 and the Ph.D. degree in control engineering from the School of Electrical and Computer Engineering, The University of Newcastle, Callaghan, NSW, Australia, in 2011.

He joined the Australian Research Council (ARC) Centre of Excellence for Complex Dynamic Systems and Control. He is currently with

the Federal University of Rio Grande do Sul, Porto Alegre, Brazil. His research interests include applied control, high precision motion control, nonlinear systems, and power converters.



Jinchuan Zheng (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees in mechatronics engineering from Shanghai Jiao Tong University, Shanghai, China, in 1999 and 2002, respectively, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2006.

He joined the Australian Research Council (ARC) Centre of Excellence for Complex Dynamic Systems and Control, School of Electrical and Computer Engineering, the University of Newcastle, Callaghan, NSW, Australia, as a Research Academic in 2005. From 2011 to 2012, he was a Staff Engineer with the Western Digital Hard Disk Drive R&D Center, Singapore. Since 2012, he has been with Swinburne University of Technology, Melbourne, VIC, Australia. He is currently an Associate Professor with the Faculty of Science, Engineering and Technology. His research interests include high precision motion control systems, electric vehicle control technology, mobile robots, and biomechatronic devices.



Minyue Fu received the bachelor's degree from the University of Science and Technology of China, Hefei, China, in 1982, and the master's and Ph.D. degrees from the University of Wisconsin-Madison, Madison, WI, USA, in 1983 and 1987, respectively, all in electrical engineering.

From 1987 to 1989, he was an Assistant Professor with the Department of Electrical and Computer Engineering, Wayne State University, USA. He joined the University of Newcastle, Australia, in 1989. He is currently a Chair Professor in Electrical Engineering with the University of Newcastle. His main research interests include control systems, signal processing, and communications.

Dr. Fu has been an Associate Editor for the IEEE TRANSACTIONS ON AUTOMATIC CONTROL, IEEE TRANSACTIONS ON SIGNAL PROCESSING, *Automatica* and *Journal of Optimization and Engineering*. He is a Fellow of IEEE, Fellow of Engineers Australia, and Fellow of Chinese Association of Automation.



Jeferson Vieira Flores was born in Carlos Barbosa, Rio Grande do Sul, Brasil. He received the B.S., M.Sc., and Ph.D. degrees in electrical engineering from the Federal University of Rio Grande do Sul (UFRGS), Porto Alegre, Brazil, in 2005, 2008, and 2012, respectively.

From 2012 to 2014, he was an Associate Professor with the Department of Electrical Engineering, PUCRS. He joined the Department of Electrical Engineering, UFRGS, in 2013, where he is currently an Adjunct Professor. His main research subjects are nonlinear control, output regulation, robust control, control of power converters, and vibration control in wind turbines.